

The Digital Acoustic Monitoring Instrument: DMON2

The DMON2 is a battery-powered underwater passive acoustic monitoring instrument specifically designed for detecting and recording marine mammal sounds. With appropriate hydrophones and firmware, the device is capable of continuous or duty-cycled acquisition and processing of sound over the full frequency range of whale vocalizations: 10 Hz - 160 kHz. Sound recordings can be stored in the built-in non-volatile flash memory for later analysis. Data are offloaded from the DMON2 via USB. The device can be used as a stand-alone recorder or real-time detector, or can be combined with an external GPS and radio telemetry as part of a marine mammal monitoring installation.

Specifications:

- DMON2 hydrophone head includes 3 Navy type II end-capped ceramic cylinders:
 - LF (low frequency): -203 dB re V/ μ Pa rms sensitivity, 10 Hz - 7.5 kHz bandwidth
 - MF (mid-frequency): -200 dB re V/ μ Pa rms sensitivity, 100 Hz - 50 kHz bandwidth
 - HF (high frequency): -206 dB re V/ μ Pa rms sensitivity, 1 kHz - 160 kHz bandwidth
- LF noise floor: 32 dB re μ Pa/ $\sqrt{\text{Hz}}$ at 10 kHz.
- 20 dB preamp gain at hydrophone and additional 4.6/13.2 dB gain (selectable) at audio board
- 16-bit A/D conversion with 1.5 V zero-to-peak voltage
- On-board, fixed-point, digital signal processor (DSP) capable of running sound detection / classification / compression algorithms, including the low-frequency detection and classification system (LFDCS)
- Low power: < 200 mW
- 64 GB non-volatile flash memory
- USB full-speed (12Mbps) interface for data offload
- Serial ports supporting RS232/485 communication
- Temperature compensated timing crystal with less than +/- 2ppm drift over 0-30°C.
- Rechargeable Li-Ion battery
- Optional external power input (7.5-36 V input range)
- Fully integrated into Teledyne Webb Research Slocum gliders
- Can be supplied with recording firmware or LFDCS